Replication package

"Monetary Policy Transmission through Cross-Selling Banks"

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This README follows the template of the Social Science Data Editors (https://social-science-data-editors.github.io/template_README/).

1. Overview

The code in this replication package reproduces every table and figure of the paper (except Figure 6, which uses confidential ECB supervisory survey data) from the analysis panels described below, using Stata. The micro data are confidential Norwegian administrative registers and cannot be redistributed; the package therefore ships a **pseudo data set** with the exact structure of the analysis data (identifiers, variables, panel dimensions, sign patterns), generated by `code/1_simulate_data.do`, so that every program can be run end to end. Numbers produced on the pseudo data are **not** the paper's numbers. Running the package on the confidential data requires only placing the input files in `data/raw/` and switching one flag. One master file, `code/0_master.do`, runs everything; on the pseudo data this takes about one minute.

Starting point of the code ("semi-raw" data)

The package starts from the bank–household relationship × year analysis panel (`data_analysis_final_full.dta`) and a handful of small auxiliary files, not from the raw registers. The raw registers (annual tax returns and the bank-reported deposit and loan accounts of every Norwegian resident, 2004–2018, plus the ORBOF bank reports) are accessible only inside the secure environment of Statistics Norway, and the scripts that merge them and build the relationship panel are specific to that environment's file layout and identifiers and cannot be run outside it. Every step that defines an analysis variable, a sample, or an outlier rule used in the paper — the cross-selling measures (PrB and the intensive-margin predictions), the bank–municipality–year collapse, the deposit-spread trimming, growth rates, winsorization, fixed-effect samples, the bank-level betas — is contained in this package and documented in the code.

2. Data Availability and Provenance Statements

2.1 Statement about rights

☒ I certify that the author(s) of the manuscript have legitimate access to and permission to use the data used in this manuscript.

☐ I certify that the author(s) of the manuscript have documented permission to redistribute/publish the data contained within this replication package. The administrative micro data cannot be redistributed (see below); the public series (policy rate, benchmark yields, monetary-policy surprises) are obtainable from the sources listed below.

2.2 Summary of availability

☐ All data are publicly available.

☒ Some data cannot be made publicly available.

☐ No data can be made publicly available.

2.3 Details on each data source

Data	Source	Used for	Provided in package?
Bank–household deposit and loan relationships (balances, interest earned/paid, effective rates), 2004–2018	Norwegian Tax Administration (Skatteetaten), third-party reports by banks, accessed through Statistics Norway	All exhibits	No — confidential; pseudo data
Individual tax records (income, wealth, demographics, education, child benefit, residence), 2004–2018	Norwegian Tax Administration and Statistics Norway population registers	PrB estimation, controls, municipality characteristics	No — confidential; pseudo data
Bank balance-sheet and income-statement data (ORBOF reporting database)	Statistics Norway (ORBOF is reported jointly to Finanstilsynet, Norges Bank and Statistics Norway)	Table 10 sample splits; firm-loan files	No — confidential; pseudo data
Norges Bank key policy rate (annual averages of daily rates)	Norges Bank, https://www.norges-bank.no/en/topics/Statistics/Interest-rates/	All exhibits	Pseudo version generated by `1_simulate_data.do`; for a real-data run place the series in `data/raw/policy_rate.dta`
3-month money-market rate (NIBOR) and 10-year government bond yield, annual averages	Norges Bank (government bond yields); OECD Main Economic Indicators (3-month interbank rate and 10-year yield for Norway, https://data-explorer.oecd.org)	Figure 2	Pseudo version generated by `1_simulate_data.do`; for a real-data run place the series in `data/raw/benchmark_rates.dta`
Narrative monetary-policy surprises for Norway	Holm, Paul and Tischbirek (2021), replication files of the article (Journal of Political Economy 129(10)), aggregated to annual sums	Table A8	Pseudo version generated by `1_simulate_data.do`; for a real-data run place the series in `data/raw/mpshocks_annual.dta`
ECB SSM supervisory survey on cross-selling, 2023	European Central Bank, confidential supervisory data	Figure 6 only	No — not reproducible; the figure is descriptive and no estimate in the paper depends on it

Access to the confidential data. The micro data were obtained from the Norwegian Tax Authorities and Statistics Norway under a data-use agreement with the authors' institutions, and were processed inside Statistics Norway's secure research environment. Interested researchers can contact these institutions to obtain approval for data usage (Statistics Norway: "Data for research", https://www.ssb.no/en/data-til-forskning). Access is granted to accredited research institutions for specified projects; the authors are not permitted to share the data, and observation identifiers are pseudonymised by the provider.

2.4 Dataset list

Data file (in `data/raw/`)	Unit	Variables required by the code	Origin
`data_analysis_final_full.dta`	relationship × year	`relid hhid bankid muniid year prate depositvol ln_deposits`	data-construction step inside Statistics Norway

Data file (in `data/raw/`)	Unit	Variables required by the code	Origin
		`deprate depspread_w0100_tr595 loanvol debt interest_debt loanspread loanspread_on age AgedSub30 parent retired ln_inc ln_wealth number_banks_hhid hhi_deposit hhi_loan existing_dep futureborrower converted meandebt lnmeandebt maxdebt maxloanvol`	(confidential)
`mshares_and_pshares.dta`	bank × municipality × year	`bankid muniid year mshare_dep`	deposit market shares computed from the relationship data (confidential)
`muni_characteristics.dta`	bank × municipality × year	`bankid muniid year total_income gross_wealth gross_fin_assets total_deposits educ move`	bank × municipality × year means of household observables (confidential)
`bankstrength.dta`	bank × year	`bankid year ta ln_ta profits_ta`	ORBOF bank reports (confidential)
`firmloans_bankmuni.dta`	bank × municipality × year	`bankid` (string; numeric id in positions 4–10) `muniid year ln_firmloans_bmy firmloangrowthld_w199`	NFC loan reports (confidential)
`firmloans_banksector_nace3.dta`	bank × NACE3 × year	`bank_id nace3digits year ln_firmloans_bsy firmloangrowthld_w199`	NFC loan reports (confidential)
`firmloans_banksector_nace5.dta`	bank × NACE5 × year	`bank_id F_nace year ln_firmloans_bsy firmloangrowthld_w199`	NFC loan reports (confidential)
`policy_rate.dta`	year (2003–2018)	`year prate2 d_prate2`	Norges Bank (public)
`benchmark_rates.dta`	year	`year nibor3m y12m y3y y5y y10y`	Norges Bank / OECD MEI (public)
`mpshocks_annual.dta`	year	`year mpshock`	Holm, Paul and Tischbirek (2021) (public)

With `global simulated 1` (default) all ten files are generated in `data/simulated/` by `1_simulate_data.do` and read from there; for a real-data run all ten are placed in `data/raw/`. `firmloangrowthld_w199` must arrive as a fraction (the code multiplies by 100 and stops with a scaling error otherwise).

3. Computational requirements

3.1 Software

- Stata 16 or later (the code sets `version 16`). The Stata flavour, version and operating system of each run are written to `logs/environment.txt` by `0_master.do`, together with the version line of each community-contributed package.

- Community-contributed Stata packages, installed with `ssc install <name>, replace`: `reghdfe`, `ftools`, `estout` (provides `esttab`, `eststo`, `estpost`, `estadd`), `winsor2`, `binscatter`. `config.do` checks that they are installed and stops with a message if not.
- No other software is required. Tables are written directly as LaTeX and figures as PNG.

3.2 Controlled randomness

☒ Random seed is set at line 35 of `code/1_simulate_data.do` (`set seed 20261021`). No other program uses random numbers.

3.3 Memory, run time and storage

- Pseudo data (50 banks, 120 municipalities, 30,000 households): about one minute on a standard laptop; less than 100 MB of disk.
- Confidential data (about 180 banks, 430 municipalities, 45 million relationship-years): the relationship-level steps (2 and 5) dominate; a full run takes several hours.

4. Description of programs/code

```
replication_package/
├── README.docx / README.pdf
├── LICENSE.txt
├── code/
│   ├── 0_master.do                   runs everything; writes logs/environment.txt
│   ├── config.do                     paths, file names, helper programs, $simulated
│   ├── 1_simulate_data.do            pseudo data (only when $simulated == 1)
│   ├── 2_construct_prb_and_bmy.do    PrB, intensive-margin predictions, BMY panel
│   ├── 3_setup_bmy_panel.do          common analysis panel
│   ├── 4_tables_pricing.do           Tables 3-6, 10, A1-A3, A6-A8; T1 Panel B
│   ├── 5_tables_relationship_level.do  Tables 2, 9, A5; Fig. 5; T1 Panel A
│   ├── 6_tables_loangrowth.do        Tables 7, 8, A4; T1 Panels C1-C3
│   ├── 7_figures.do                  Figures 1-4, A1; T1 Panel D
│   ├── 8_assemble_table1.do          Table 1 wrapper
│   └── 9_text_numbers.do             numbers quoted only in the text
├── data/
│   ├── raw/        <- input files for a real-data run go here
│   ├── simulated/  <- written by 1_simulate_data.do
│   └── work/       <- intermediate files written by the pipeline
├── output/
│   ├── tables/     <- .tex files named as in manuscript.tex
│   └── figures/    <- .png files named as in manuscript.tex
└── logs/           <- replication_run.log, environment.txt
```

- `config.do` defines three helper programs used throughout: `write_table` (esttab wrapper producing a self-contained `table` environment with the paper's legend), `write_sumstats_panel` (one panel of Table 1) and `label_bmy_vars` (variable labels).
- Steps 2–3 build the analysis panels once (`data/work/`); steps 4–9 read them. Each of `code/1_…` to `code/9_…` can be run on its own, in order: every script first checks that the helper programs from `config.do` are loaded and re-runs `config.do` if not. Avoid `clear all` between steps, as it drops those programs.

- The intermediate files `BMY_unweighted_withconv.dta`, `overp50s.dta` and `treatments_banklevel_weighted_conv.dta`, are rebuilt in step 2 from the relationship panel, so they are not inputs.

4.1 License for code

The code is licensed under the MIT license. See `LICENSE.txt` for details.

5. Instructions to replicators

1. Unzip the package; open Stata and `cd` to the package folder (the one containing `code/`). `0_master.do` sets `global root` to the current working directory; edit that line if you run it from elsewhere.
2. In `code/config.do` set `global simulated 1` to run on the pseudo data (default) or `global simulated 0` to run on the data placed in `data/raw/`.
3. `do "code/0_master.do"`. The log is written to `logs/replication_run.log`.
4. Copy `output/tables` and `output/figures` next to `manuscript.tex` and compile (`manuscript.tex` defines the `\tablelegend` and `\sym` macros the tables use).

6. List of tables, figures and programs

Table numbers follow the order of the `\input` lines in `manuscript.tex`; the output file names are the manuscript's file names, so no `\input` line changes.

Exhibit	Output file	Script
Table 1 (summary statistics)	`t1_sumstats.tex` (+ `t1_panelA..D.tex`)	4, 5, 6, 7, 8
Table 2 (LPM)	`t2_lpm.tex`	5
Table 3 (core pricing)	`t3_core.tex`	4
Table 4 (intensive margin)	`t4_intensive.tex`	4
Table 5 (other banks' PrB)	`a5_other.tex`	4
Table 6 (deposit growth)	`t5_depgrowth.tex`	4
Table 7 (loan growth, bank–municipality)	`t6_lg_bmy.tex`	6
Table 8 (loan growth, NACE3)	`t7_lg_nace3.tex`	6
Table 9 (flow profits)	`t8_flow.tex`	5
Table 10 (low rates)	`t9_lowrates.tex`	4
Table A1 (Altonji–Oster, residualized)	`a4_ao.tex` (+ `a4_ao_variance_decomposition.txt`)	4
Table A2 (leads/lags)	`a2_leads.tex`	4
Table A3 (loan HHI)	`a7_loanhhi.tex`	4
Table A4 (NACE5)	`a8_lg_nace5.tex`	6
Table A5 (NPL premium)	`a9_npl.tex`	5
Table A6 (levels)	`a11_levels.tex`	4
Table A7 (weighted)	`a12_weighted.tex`	4

Exhibit	Output file	Script
Table A8 (MP surprises)	`a13_mps.tex`	4
Figure 1	`figure1_spreads_v2.png` (+ `figure1_annual_series.csv`)	7
Figure 2	`figure2_benchmarks_v2.png`	7
Figure 3	`figure3_hist_conv.png`	7
Figure 4	`figure4_panelA_conv_v2.png` (Panel B also produced)	7
Figure 5	`figure5a_loanspread_sameyear.png`, `figure5b_loanspread_onboarding.png`	5
Figure 6	`figure6_ecb_crossselling.png`	—
Figure A1	`figureA1_hhi_loan_vs_deposit.png`	7
Sec. 5.7 descriptives (negative spreads, alternative benchmarks)	`lowrate_descriptives.txt`	7
Sec. 3.3 / 4.1 / 4.3 descriptives (policy-rate range, within-bank-year SD of deposit rates, municipalities per bank, loan-growth sample size)	`text_numbers.txt`	9

6.1 Notes on individual exhibits

- **Table 10, column 1** uses bank-year fixed effects and two-way clustering by bank and year, like columns 2–5.
- **Figures 1 and 2** plot deposit-volume-weighted annual means over all bank-municipality-year rows of the panel; the series is also written to `figure1_annual_series.csv`.
- **Figure 5** shows binned scatters of the loan spread at onboarding over the policy rate in the same year and over the policy rate in the relationship's first observed year, on relationship-years with a positive loan balance.
- **Figure 6** is based on confidential ECB supervisory survey data and is not reproduced by this package; it is descriptive and no estimate in the paper depends on it.
- **Pseudo data simplifications.** The simulated `loanspread` equals `loanspread_on`, and the simulated `policy_rate.dta` contains approximate annual averages of the Norges Bank key rate. Neither affects the code path.

7. References

Holm, M. B., Paul, P., Tischbirek, A., 2021. The transmission of monetary policy under the microscope. Journal of Political Economy 129 (10), 2861–2904.

Norges Bank. Interest rates (key policy rate; government bond yields). https://www.norges-bank.no/en/topics/Statistics/Interest-rates/

OECD. Main Economic Indicators: short-term and long-term interest rates, Norway. https://data-explorer.oecd.org

Statistics Norway. Data for research. https://www.ssb.no/en/data-til-forskning